

Getting "NotImplementedError: Could not run 'torchvision::nms' with arguments from CUDA backend" despite having all necessary libraries and imports

Asked 2 years, 1 month ago Modified 8 months ago Viewed 41k times



The full error:

16



```
NotImplementedError: Could not run 'torchvision::nms' with arguments from the 'CUDA' backend. This could be because the operator doesn't exist for this backend, or was omitted during the selective/custom build process (if using custom build). If you are a Facebook employee using PyTorch on mobile, please visit https://fburl.com/ptmfixes for possible resolutions. 'torchvision::nms' is only available for these backends: [CPU, QuantizedCPU, BackendSelect, Python, FuncTorchDynamicLayerBackMode, Functionalize, Named, Conjugate, Negative, ZeroTensor, ADInplaceOrView, AutogradOther, AutogradCPU, AutogradCUDA, AutogradXLA, AutogradMPS, AutogradXPU, AutogradHPU, AutogradLazy, Tracer, AutocastCPU, AutocastCUDA, FuncTorchBatched, FuncTorchVmapMode, Batched, VmapMode, FuncTorchGradWrapper, PythonTLSSnapshot, FuncTorchDynamicLayerFrontMode, PythonDispatcher].
```

I get this when attempting to train a YOLOv8 model on a Windows 11 machine, everything works for the first epoch then this occurs.

I also get this error immediately after the first epoch ends but I don't think it is relevant.

```
Error executing job with overrides: ['task=detect', 'mode=train', 'model=yolov8n.pt', 'data=custom.yaml', 'epochs=300', 'imgsz=160', 'workers=8', 'batch=4']
```

I was trying to train a YOLOv8 image detection model utilizing CUDA GPU.

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edited Jan 12, 2023 at 23:55

 **talonmies**
72.3k ● 35 ● 202 ● 287

asked Jan 12, 2023 at 22:50

 **Vincent Casey**
730 ● 1 ● 7 ● 16

7 Answers

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If you get here after updating the new drivers (CUDA 12.4), you have to install a PyTorch that is still in a previous version

3

```
pip uninstall torch torchvision torchaudio
```

pip3 install --pre torch torchvision torchaudio --index-url
https://download.pytorch.org/whl/nightly/cu124



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edited Jun 13, 2024 at 7:45

answered Jun 12, 2024 at 8:56



Vitalizzare

7,240 ● 10 ● 20 ● 43



Vendido por

31 ● 1

1

I solved the problem following Pamudu suggestion. First I checked my CUDA version by running `nvidia-smi` on my Windows terminal. Then I went to the PyTorch web site and found the right command to install the version that matches my CUDA version. In my case the `nvidia-smi` reported version 12.2, the command below did the trick.

```
pip3 install torch torchvision torchaudio --index-url  
https://download.pytorch.org/whl/cu121
```



As you probably has already installed torch so must uninstall everything first.

```
pip uninstall torch torchvision torchaudio
```

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answered Apr 29, 2024 at 14:50



Ángel Tena

11 ● 3

6

I had this issue and had no idea how to fix it. I then installed the PyTorch CUDA with `pip` instead of `pip3`, and that fixed it.

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edited Apr 1, 2024 at 3:04

answered Apr 1, 2024 at 1:56



L Tyrone

6,970 ● 23 ● 28 ● 41



Michael Peccia

61 ● 1 ● 1



This doesn't sound like a reliable method. – Chris Fu Apr 1, 2024 at 7:08

1



This fixed it for me too, no idea why, didn't even know there was a difference.. My only guess is maybe `pip3` is using the system's python interpreter and `pip` is getting redirected to the venv's interpreter, which are different python versions. – tenten8401 Apr 3, 2024 at 0:53



5



This issue arises because you have installed a PyTorch version that is not compatible with the CUDA version (Compute Platform). You can identify your CUDA version using the `nvidia-smi` command.

For example, if your CUDA version is 12, but you have installed a PyTorch version that is compatible with CUDA 12.1. To resolve this, uninstall PyTorch and reinstall it to match your CUDA version (computer platform). [Find the relevant PyTorch version](#)

If your CUDA version is 12, and you cannot find a PyTorch version compatible with that, install a PyTorch version compatible with a CUDA version less than 12, such as 11.8. No issues will arise in this case.

Note that you cannot install a PyTorch version with a higher CUDA version than your current platform.

If you have installed the CPU version of PyTorch, this problem should not occur.

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answered Feb 17, 2024 at 2:32

 **Pamudu Ranasinghe**
91 ● 1 ● 3



Vote



I don't know what causes this error but I know installing the proper version of torchvision and cuda didn't fix it. The way I solved is by uninstalling all of my packages and then reinstalling. Works just fine now. In hindsight, it may have been better to just update all my packages, but problem fixed none the less.

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answered Apr 15, 2023 at 8:48

 **Legendary Master**
27 ● 4



34



This error occurs when you have Torch and Torchaudio for CUDA but not Torchvision for CUDA installed.

Uninstall Torch and Torchvision, I used pip:

```
pip uninstall torch torchvision
```

Then go [here](#) to install the proper versions of both using the nice interface. I got the following command:

```
pip install torch torchvision torchaudio --extra-index-url
https://download.pytorch.org/whl/cu116
```

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answered Jan 13, 2023 at 17:36

 **Vincent Casey**
730 ● 1 ● 7 ● 16

▲ the same problem. After exec you commands I have: `ValueError: Invalid CUDA 'device=0' requested.`
 ▼ Use `'device=cpu'` or pass valid CUDA device(s) if available, i.e. `'device=0'` or `'device=0,1,2,3'` for Multi-GPU. `torch.cuda.is_available(): False torch.cuda.device_count(): 0 os.environ['CUDA_VISIBLE_DEVICES']: None` See <https://pytorch.org/get-started/locally/> for

 **This post is hidden.** It was [deleted](#) last year by [Stephen Rauch](#) ♦.

-2

Same for me. Reinstalling of torch helped. Thanks

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answered Feb 2, 2024 at 17:32



Rafal

479 ● 1 ● 4 ● 10

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duplicate of stackoverflow.com/a/76021232 – [karel](#) Feb 2, 2024 at 17:57



Please don't add "thank you" as an answer. Instead, vote up the answers that you find helpful. - [From Review](#)



– [Rich](#) Feb 2, 2024 at 18:26

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